

Epigenetic Landscape Dynamics Underlying Stochastic Cell and Individual Fate Determination

While genetic variation frequently drives phenotypic diversity, genetically identical individuals can still exhibit markedly distinct traits. These differences may arise from rare somatic mutations, environmental fluctuations, or stochastic events during development, often resulting in divergent epigenetic profiles. Understanding the relative contributions of these factors is essential for elucidating the complex mechanisms underlying human cellular and individual phenotypic divergence.

In this talk, I will introduce our time-course transcriptome analysis on the nine-banded armadillo (*Dasypus novemcinctus*), which produce litters of genetically identical quadruplets. Analyzing the transcriptome profiles within the siblings indicated the existence of identity signatures under tight control of genetic and environmental factors. Finally, I will discuss our ongoing work, where the transcriptome and epigenetic variability is further assessed on large-scale omics data including hundreds of human pluripotent stem cell lines.